

Matrix discrepancy and optimization

The sharp Lovász-theta constant for dense random graphs

Difficulty: challenging **Importance:** broadly interesting**Status:** Solution claimed

Rating rationale: A sharp expectation constant requires new random SDP analysis beyond order bounds; it connects matrix optimization with probabilistic combinatorics.

The difficulty and importance ratings above are historical ratings of the original open target.

Full-resolution claim — checked 6 October 2026

Aaron Potechin and Jeff Xu, *Sharp Lovász-Theta Bounds on Random Graphs*, arXiv:2609.30064v1, first submitted 24 September 2026. In §1.1, **Theorem 1.2 and Corollary 1.3** (printed p. 3 of the [full manuscript](#)), the authors claim that $\vartheta(G_n)/\sqrt{n} \rightarrow 1$ in probability. Appendix F (printed pp. 76–77) identifies their formulation with the trace-normalized SDP used in this entry. The v1 corollary prints an exact high-probability lower bound $\vartheta(G_n) \geq \sqrt{n}$. Coauthor Jeff Xu confirmed in correspondence supplied by the maintainer on 8 October 2026 that its intended lower bound is $(1 - o_n(1))\sqrt{n}$. He also confirmed that Claim 4.15’s proof should have a logarithm, rather than n , inside the disputed square root. These corrections leave the leading-constant claim unchanged; a corrected arXiv version was not yet available on 8 October 2026. The [official FOCS 2026 accepted-paper list](#) includes this paper; acceptance is recorded separately from publication in the proceedings.

Correspondence with the expectation target. The claimed probabilistic limit implies the original expectation limit by the following elementary uniform-integrability argument. Let A_n be the ordinary zero-one adjacency matrix and put $B_n = J - I - 2A_n$. For every matrix X feasible in the original SDP, the edge constraints give

$$\langle J, X \rangle = \langle I + B_n, X \rangle \leq 1 + \|B_n\|_2, \quad 0 \leq \frac{\vartheta(G_n)}{\sqrt{n}} \leq \frac{1 + \|B_n\|_2}{\sqrt{n}}.$$

The independent upper-triangular entries of B_n are uniform signs. For a fixed real unit vector x , the sum $x^T B_n x = 2 \sum_{i < j} (B_n)_{ij} x_i x_j$ has squared coefficient sum at most two. The elementary exponential-moment bound for a sum of independent signs therefore gives $\Pr\{|x^T B_n x| > t\} \leq 2e^{-t^2/4}$. A $1/4$ -net of the real unit sphere with at most 9^n points controls the norm of a symmetric matrix within a factor of two. Consequently,

$$\Pr\{\|B_n\|_2 > s\} \leq \min\{1, 2 \cdot 9^n e^{-s^2/16}\}, \quad \mathbb{E}\|B_n\|_2^2 \leq 16(n \log 9 + \log 2 + 1).$$

The moment bound follows by integrating this tail bound. Thus $\sup_n \mathbb{E}[(\vartheta(G_n)/\sqrt{n})^2] < \infty$, so the normalized Lovász numbers are uniformly integrable. Theorem 1.2’s high-probability upper bound therefore gives $\limsup_n \mathbb{E}\vartheta(G_n)/\sqrt{n} \leq 1$. For the lower bound, the deterministic inequality $\vartheta(G)\vartheta(\bar{G}) \geq n$ and the fact that G_n and $\bar{G}_n$ have the same distribution give, by the arithmetic-geometric mean inequality, $\mathbb{E}\vartheta(G_n) \geq \sqrt{n}$ for every n . Together these establish the original expectation limit, conditional on Theorem 1.2. The argument preserves the original SDP, graph ensemble and quantifiers; it does not replace the expectation statement with a probabilistic one.

Evidence level, 8 October 2026: an [independent AI-agent review](#) checked the theorem’s match to the expectation target, the bridge above, and the witness construction through its main dependency chain. It could not independently certify the decisive graph-matrix norm, product and correction estimates (Theorem 3.27/Corollary 3.28 and Theorems 4.7 and 4.17). The follow-up review found a concrete counterexample to auxiliary Theorem 3.24’s printed $O(1)$ remainder for general fixed mixed trace moments. The later main estimates use weaker bounds compatible with that example, so it does not by

itself refute Theorem 1.2, but the published proof needs clarification. No MD-01 Lean formalization or verification record was found in the checked public sources; the paper’s “Formal Verification” appendix is a written argument. The status therefore remains **Solution claimed**; no Lean verification or external human peer review is asserted. The original mathematical statement follows unchanged.

Original problem statement

Let G_n be the simple random graph on n labelled vertices in which each unordered pair is an edge independently with probability $1/2$. For a graph G , define its Lovász number by

$$\vartheta(G) = \max\{\langle J, X \rangle : X \in \mathbb{R}^{n \times n}, X = X^T \geq 0, \operatorname{tr} X = 1, X_{ij} = 0 \text{ for } \{i, j\} \in E(G)\},$$

where J is the all-ones matrix and $\langle J, X \rangle = \sum_{i,j} X_{ij}$.

Conjecture. $\lim_{n \rightarrow \infty} \mathbb{E} \vartheta(G_n) / \sqrt{n} = 1$.

This asks for the leading constant of a random semidefinite matrix optimization problem. It is an expectation statement; convergence in probability is not substituted for it.

References

1. A. S. Bandeira, D. Dmitriev, K. Lucca, P. Nizić-Nikolac, and A. Rödder, *Randomstrasse101: Open Problems of 2025*, Entry 9, Conjecture 17 and the preceding SDP definition. [Archived paper](#).
2. A. Potechin and J. Xu, *Sharp Lovász-Theta Bounds on Random Graphs*, arXiv:2609.30064v1, 24 September 2026, §1.1, Theorem 1.2 and Corollary 1.3, printed p. 3; accepted for FOCS 2026.

Status check — 2026-09-10

The archived 2026 paper retains the conjecture. Targeted searches for the Lovász number of dense Erdős–Rényi graphs and a sharp leading constant through September 2026 found no resolution. The adjacent circulant problem uses a different random matrix ensemble and is recorded separately.

Audit update (2026-09-10): Rechecked the 2026 Conjecture 17 and searched for a sharp dense-random-graph theta constant. No resolution of the expectation limit was located. This is a bounded literature check, not a proof that no solution exists.